



CHAPTER

1

Solutions

Sr. No.	Concept	Formulas	Other information
1.	Mass percentage (w/w) %	Mass % of solute $= \frac{\text{Weight of the solute (g)}}{\text{Weight of the solution (g)}} \times 100$	NA
2.	Volume percentage (V/V) %	Volume % of solute $= \frac{\text{Vol. of solute (mL)}}{\text{Vol. of solution (mL)}} \times 100$	NA
3.	Mass by volume percentage (w/V) %	(w/V %) of solute $= \frac{\text{Weight of solute (g)}}{\text{Volume of solution (mL)}} \times 100$	NA
4.	Molarity (M)	Molarity (M) $= \frac{\text{Number of moles of solute (mol)}}{\text{Volume of solution in liters}}$	Here $\text{Mole} = \frac{\text{Mass of the solute (in g)}}{\text{Molecular mass of the solute (g/mol)}}$
5.	Molality (m)	Molality (m) $= \frac{\text{Number of moles of solute (mol)}}{\text{Mass of solvent (kg)}}$ $= \frac{\text{Moles of the solute (mol)}}{\text{Mass of solvent (g)}} \times 1000$	Here $\text{Mole} = \frac{\text{Mass of the solute (in g)}}{\text{Molecular mass of the solute (g/mol)}}$
6.	Mole fraction (χ)	Mole fraction of solute in solution $\chi_2 = \frac{n_2}{n_1 + n_2} = \frac{\frac{w_2}{M_2}}{\frac{w_1}{M_1} + \frac{w_2}{M_2}}$ Mole fraction of solvent in solution $\chi_1 = \frac{n_1}{n_1 + n_2} = \frac{\frac{w_1}{M_1}}{\frac{w_1}{M_1} + \frac{w_2}{M_2}}$	Here, w_1 and M_1 are weight and molar mass of solvent respectively. w_2 and M_2 are weight and molar mass of the solute respectively. n_1 is the mole of solvent. n_2 is the mole of solute.

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7.	Relation between molarity (M) and molality (m)	$m = \frac{1000M}{1000d - MM_1} \text{ (mol kg}^{-1}\text{)}$	Here. m is molality of the solution M is molarity of the solution M ₁ is molar mass of solute d is density of the solution (g/mL)
8.	Relation between molarity (M) and mole fraction (χ)	$M = \frac{\chi_2 \times 1000 \times d}{\chi_1 m_1 + \chi_2 m_2} \text{ (mol L}^{-1}\text{)}$	Here. m ₁ and m ₂ are molar masses of solvent and solute respectively d is density of solution in g/mL M is molarity of the solution χ ₁ is mole fraction of solvent χ ₂ is mole fraction of solute
9.	Parts per million (ppm)	Parts per million (ppm) $= \frac{\text{Mass of solute (g)}}{\text{Mass of solution (g)}} \times 10^6$ or $= \frac{\text{Volume of solute (mL)}}{\text{Volume of solution (mL)}} \times 10^6$	NA
10.	Composition of vapour phase (Dalton's law of partial pressures)	$p_1 = y_1 p_{\text{total}}$ $p_2 = y_2 p_{\text{total}}$	Here, p ₁ and p ₂ be the partial vapour pressures of the two components 1 and 2 respectively. These partial pressures are related to the mole fractions y ₁ and y ₂ of the two components 1 and 2 respectively. p _{total} is the total vapour pressure.
11.	Henry's law:	$p = K_H \chi$	Here K _H is called Henry's constant χ is the mole fraction of gas. p is the pressure of the gas above the solution.

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12.	Raoult's law (a) Raoult's law, for solution containing volatile components 1 and 2 (b) Raoult's law for non-volatile solute containing solution	$p_1 = p_1^0 \chi_1 \text{ and } p_2 = p_2^0 \chi_2$ $p_{\text{total}} = \chi_1 p_1^0 + \chi_2 p_2^0$ $= (1 - \chi_2) p_1^0 + \chi_2 p_2^0$ $= p_1^0 + (p_2^0 - p_1^0) \chi_2$ $\frac{p^0 - p_s}{p^0} = \frac{n_2}{n_1 + n_2} = \frac{n_2}{n_1}$ $= \frac{\frac{w_2}{M_2}}{\frac{w_1}{M_1} + \frac{w_2}{M_2}} \approx \frac{\frac{w_2}{M_2}}{\frac{w_1}{M_1}} \text{ (if solution is dilute)}$	Here, p_{total} is the total vapour pressure. χ_1 and χ_2 are the mole fraction of the components 1 and 2 respectively. p_2^0 is the vapour pressure of component 2 in pure state. p_1^0 is vapour pressure of component 1 in the pure state. p^0 is vapour pressure of pure component and p_s is vapour pressure of solution. n_1 and n_2 are the mole of components 1 and 2.
13.	Colligative Properties: (a) Relative lowering of V.P. For dilute solutions $n_2 \ll n_1$. (b) Elevation in Boiling Point	$\frac{p^0 - p_s}{p^0} = \frac{n_2}{n_1 + n_2} \left(\text{since } \chi_2 = \frac{n_2}{n_1 + n_2} \right)$ $\frac{p^0 - p_s}{p^0} = \frac{n_2}{n_1}$ $\Delta T_b = (T_b - T_b^0) = K_b \times m$ $\Delta T_b = \frac{K_b \times 1000 \times w_2}{M_2 \times w_1}$ $K_b = \frac{R \times M_1 \times (T_b^0)^2}{1000 \times \Delta_{\text{vap}} H}$	Here, n_1 - Number of moles of solvent n_2 - Number of moles of non-volatile solute p_s - Vapour pressure of the solution p^0 - Vapour pressure of component in the pure state. χ_2 - Mole fraction of solute m - Molality of the solution. K_b - Boiling Point Elevation Constant. T_b^0 - Boiling point of the pure solvent in kelvin. T_b - Boiling point of solution R - gas constant M_1 - Molar mass of the solvent $\Delta_{\text{vap}} H$ - Enthalpy of vapourisation w_2 - Weight of solute(g) M_2 - Molar mass of solute M_1 - Molar mass of the solvent

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	(c) Depression in Freezing Point	$\Delta T_f = T_f^0 - T_f = K_f \times m$ $\Delta T_f = \frac{K_f \times w_2 \times 1000}{M_2 \times w_1}$ $K_f = \frac{R \times M_1 \times (T_f^0)^2}{1000 \times \Delta_{\text{fus}} H}$	Here, T_f - Freezing point of solution K_f - Molal depression constant T_f^0 - Freezing point pure solvent in kelvin $\Delta_{\text{fus}} H$ - Enthalpy of fusion m - Molality of the solution. R - gas constant w_2 - Weight of solute(g) M_2 - Molar mass of solute π - Osmotic pressure C - Molarity of the solution T - Temperature (Kelvin) V - Volume of a solution (L) n_2 - Moles of solute.
	(d) Osmotic pressure	$\pi = C \times R \times T \quad \pi = (n_2 / V)RT$ $\pi V = \frac{w_2 RT}{M_2}$	
14.	Van't Hoff factor (i):	$i = \frac{\text{Observed value of colligative property}}{\text{Calculated value of colligative property}}$ $= \frac{\text{Calculated molar mass}}{\text{Observed molar mass}} = \frac{M_{\text{cal.}}}{M_{\text{obs.}}}$ $\left(\because \text{Molar mass} \propto \frac{1}{\text{Colligative property}} \right)$	Here, n is number of ions after association or dissociation
	(a) For an electrolyte $A_x B_y$ undergoing dissociation with degree of dissociation (α).	$i = 1 + (n - 1)\alpha \text{ or } \alpha = \frac{i - 1}{n - 1}$	
	(b) For a solute A undergoing association	$i = 1 - \alpha + \frac{\alpha}{n} \text{ or } \alpha = \frac{1 - i}{1 - \frac{1}{n}}$	

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15.	Colligative Properties with Van't Hoff factor (i):		
	(a) Relative lowering of V.P. For dilute solutions $n_2 \ll n_1$.	$\frac{p^0 - p_s}{p^0} = i \times \frac{n_2}{n_1 + n_2} \left(\text{since } \chi_2 = \frac{n_2}{n_1 + n_2} \right)$	
	(b) Elevation in Boiling Point	$\Delta T_b = (T_b - T_b^0) = i \times K_b \times m$	NA
	(c) Depression in Freezing Point	$\Delta T_f = T_f^0 - T_f = i \times K_f \times m$	
	(d) Osmotic pressure	$\pi = i \times C \times R \times T$	